

BUSINESS REQUIREMENTS DOCUMENT

Cafeteria Sales Data Analysis Project

Data Cleaning, Exploratory Analysis & Strategic Insights

Field	Details
Document Type	Business / Functional Requirements Document (BRD)
Project Name	Cafeteria Sales Data Analysis
Data Period	1 January 2023 – 31 December 2023
Dataset Size	10,000 transactions
Prepared For	Cafeteria Management / Stakeholders
Version	1.0
Status	Draft for Review

1. Introduction

1.1 Purpose of the Document

This document defines the business and functional requirements for the Cafeteria Sales Data Analysis project. It describes the problem to be solved, the scope of work, the data requirements, the cleaning and analysis tasks to be performed, and the expected deliverables. It serves as a reference for stakeholders, data analysts, and reviewers to agree on scope and success criteria before development begins.

1.2 Background

Cafeterias generate large volumes of point-of-sale transaction data every day. In most small and mid-sized operations, this data is stored but rarely analyzed in a structured way. As a result, management lacks visibility into sales patterns, item-level performance, and demand fluctuations, which limits their ability to plan inventory, staffing, and promotions effectively.

2. Problem Statement

Cafeterias often generate large volumes of sales data daily. However, this data is frequently underutilized, leading to missed opportunities in optimizing operations and increasing profitability. The lack of systematic analysis makes it difficult for cafeteria management to identify trends, forecast demand, reduce waste, or improve revenue.

This project analyzes a dataset containing 10,000 cafeteria transactions recorded over one full year (1 January 2023 to 31 December 2023). The goal is to uncover actionable insights that can help improve business performance, with a specific focus on identifying peak sales periods, best-selling items, and revenue trends.

3. Project Objectives

The primary objectives of this project are to:

- Clean and prepare the raw transactional dataset for reliable analysis.
- Identify peak sales periods (by day, week, month, and time of day, where data allows).
- Determine best-selling and worst-selling menu items.
- Analyze revenue trends across the year and detect seasonality.
- Examine relationships (correlations) between key numeric variables such as quantity and price.
- Visualize the distribution of key columns to understand data spread and outliers.
- Translate analytical findings into practical, actionable business recommendations.

4. Scope

4.1 In Scope

- Cleaning and preprocessing of the provided 10,000-row cafeteria transaction dataset.
- Handling of missing values, invalid entries (e.g., 'UNKNOWN', 'ERROR'), and duplicate records.
- Data type standardization (dates, quantities, prices).
- Descriptive statistics, correlation analysis, and distribution plots.
- Identification of sales trends, peak periods, and top/bottom performing items.
- Delivery of a Python data-cleaning script and a Word report containing visualizations, insights, and strategic recommendations.

4.2 Out of Scope

- Predictive modeling or machine learning-based demand forecasting (may be a future phase).
- Integration with live POS systems or real-time dashboards.
- Financial auditing or verification of source data accuracy against physical records.
- Changes to the cafeteria's operational systems or menu pricing.

5. Data Requirements

5.1 Source Dataset

A single dataset containing approximately 10,000 transaction records spanning 1 January 2023 to 31 December 2023.

5.2 Expected Key Fields

Field	Description	Expected Data Type
Transaction Date	Date (and time, if available) the transaction occurred	Datetime
Transaction/Item ID	Unique identifier for the transaction or line item	Integer / String
Item Name	Name of the menu item sold	String (Categorical)
Quantity	Number of units sold in the transaction	Integer
Price per Unit	Unit price of the item	Float
Total / Revenue	Quantity x Price per unit (derived if not present)	Float
Payment Method	Mode of payment, if available	String (Categorical)

5.3 Data Quality Assumptions

- The dataset may contain missing values in one or more key fields.
- Some entries may contain placeholder or invalid text such as 'UNKNOWN' or 'ERROR' in place of numeric or categorical values.

- Duplicate transaction records may exist due to system glitches or repeated submissions.
- Column names and formats are assumed to be reasonably consistent but may require standardization (e.g., date formats, currency symbols).

6. Functional Requirements

6.1 Data Cleaning Requirements

1. Missing Values: Detect missing values in transaction date, item, quantity, and price fields; apply an appropriate handling strategy (removal, imputation, or flagging) and document the rationale.
2. Invalid / Inconsistent Entries: Identify and correct entries containing placeholder values such as 'UNKNOWN' or 'ERROR'; replace, impute, or remove as appropriate.
3. Data Type Conversion: Convert quantity to integer, price per unit to float, and transaction date/time to a proper datetime format.
4. Duplicate Removal: Identify and remove duplicate transaction records that may have resulted from system glitches or repeated entries.
5. Derived Fields: Calculate a revenue/total column (Quantity x Price per Unit) if not already present, to support trend analysis.

6.2 Analytical Requirements

6. Correlation Analysis: Compute correlations between numeric columns (e.g., quantity, price, revenue) and highlight any significant relationships.
7. Distribution Analysis: Generate plots (histograms, box plots, bar charts as appropriate) to show the distribution of key columns such as price, quantity, and item frequency.
8. Trend Analysis: Aggregate sales by day, week, and month to identify peak sales periods and seasonal patterns.
9. Item Performance: Rank items by total quantity sold and total revenue generated to identify best-sellers and underperforming items.
10. Insight Generation: Summarize key findings and translate them into concrete strategies to improve sales and revenue.

7. Non-Functional Requirements

- Accuracy: Cleaning logic must be transparent and reproducible; all transformation steps should be documented in code comments.
- Maintainability: The Python script should be modular and readable, allowing future re-use on updated datasets.
- Performance: The script should process the ~10,000-record dataset efficiently (well within a few seconds/minutes on standard hardware).
- Presentation Quality: The Word report must be clearly structured, with labeled charts and concise, business-friendly narrative.

- Reproducibility: Anyone re-running the script on the same raw dataset should obtain the same cleaned output.

8. Tools & Technologies

Category	Tool / Library
Programming Language	Python 3.x
Data Handling	pandas, NumPy
Visualization	matplotlib, seaborn
Statistical Analysis	pandas / SciPy (correlation, descriptive stats)
Documentation	Microsoft Word (.docx)
Development Environment	Jupyter Notebook / .py script

9. Deliverables

#	Deliverable	Format	Description
1	Data Cleaning Script	.py	Python script that loads the raw dataset, handles missing/invalid values, fixes data types, removes duplicates, and outputs a cleaned dataset.
2	Analysis & Insights Report	.docx	Word document containing distribution plots, correlation findings, sales trend charts, key insights, and strategic recommendations.

10. Success Criteria / Acceptance Criteria

- Cleaned dataset is free of missing, duplicate, and placeholder ('UNKNOWN'/'ERROR') values in key fields.
- All key columns have correct, analysis-ready data types.
- Report includes at least: distribution plots for price and quantity, a correlation summary, monthly/seasonal sales trend, and a best-seller ranking.
- Report presents at least 3-5 clear, actionable strategic recommendations tied directly to the data findings.
- All deliverables are reviewed and approved by cafeteria management / project stakeholders.

11. Assumptions & Constraints

11.1 Assumptions

- The provided dataset is a reasonably accurate representation of actual cafeteria sales for the stated period.
- One year of data (10,000 transactions) is sufficient to identify meaningful trends and seasonality.
- Stakeholders will be available for a brief review/sign-off of the cleaned data approach before final analysis.

11.2 Constraints

- Analysis is limited to the fields available in the source dataset; no external data sources will be merged unless explicitly requested.
- The project timeline assumes no major structural issues in the raw file (e.g., corrupted rows) beyond what is described in the problem statement.

12. Stakeholders

Role	Responsibility
Cafeteria Management	Provides raw data, reviews insights, approves final recommendations
Data Analyst / Project Team	Performs data cleaning, analysis, visualization, and reporting
Project Reviewer / Sponsor	Reviews and signs off on requirements and final deliverables

13. Indicative Timeline

Phase	Activity	Estimated Duration
1	Requirements sign-off (this document)	1 day
2	Data cleaning & preprocessing (.py script)	1-2 days
3	Exploratory analysis, correlation & distribution plots	1-2 days
4	Trend analysis, insight generation, report drafting	1-2 days
5	Review & final delivery	1 day